Adenosine: Drug information - UpToDate

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For additional information see ”Adenosine: Patient drug information” and ”Adenosine: Pediatric drug information”

For abbreviations, symbols, and age group definitions show table

Pharmacologic Category

- Antiarrhythmic Agent, Miscellaneous;
- Diagnostic Agent

Dosing: Adult

Paroxysmal supraventricular tachycardia

Paroxysmal supraventricular tachycardia:

Note: Recommended for hemodynamically stable patients who do not respond to vagal maneuvers or for hemodynamically unstable patients. Preparations should be made for synchronized direct-current cardioversion in case adenosine is ineffective. For patients whose arrhythmia persists despite successful adenosine-induced AV block, switch to an alternative therapy. Do not use in patients with a known accessory pathway that exhibits retrograde conduction; this can lead to ventricular arrhythmias (Ref).

IV: Initial: 6 mg over 1 to 2 seconds via a peripheral line, followed immediately by an NS flush; if initial dose does not terminate the arrhythmia or cause AV block within 1 to 2 minutes, administer a second dose of 12 mg using the same procedures; if second dose does not terminate the arrhythmia or cause AV block, may administer a third dose of 12 or 18 mg using the same procedures. If supraventricular tachycardia terminates then recurs, may repeat the last effective dose as needed for a total of 3 doses (Ref).

Central line administration: Initial dose should be reduced to 3 mg with subsequent doses of 6 mg, then 9 mg, if needed (Ref).

Heart transplant patients: Initial dose should be reduced to 1 mg; may increase subsequent doses up to 3 mg, if needed (Ref).

Pharmacologic cardiac stress testing, diagnostic aid

Pharmacologic cardiac stress testing, diagnostic aid: Continuous infusion: IV (via peripheral line): 140 mcg/kg/minute for 6 minutes using syringe or volumetric infusion pump; total dose: 840 mcg/kg. Thallium-201 is injected at midpoint (3 minutes) of adenosine infusion.

Fractional flow reserve testing, diagnostic aid

Fractional flow reserve testing, diagnostic aid (off- label use):

Continuous infusion: IV (via peripheral line): 140 mcg/kg/minute during testing (Ref).

Intracoronary: 40 mcg into the right coronary artery or 80 mcg into the left coronary artery; dilute dose in 10 mL of NS and administer rapidly through the guiding catheter (Ref).

Tachyarrhythmia, diagnostic aid in hemodynamically stable patients

Tachyarrhythmia, diagnostic aid in hemodynamically stable patients (off-label use):

Note: May administer to hemodynamically stable patients as a diagnostic aid when trying to decipher the type of arrhythmia (eg, ventricular tachycardia vs supraventricular tachycardia with aberrancy). Do **not** administer to hemodynamically unstable patients or for irregular or polymorphic wide complex tachycardia (Ref).

IV: Initial: 6 mg over 1 to 2 seconds via a peripheral line, followed immediately by an NS flush; if initial dose does not terminate the arrhythmia or cause AV block within 1 to 2 minutes, administer a second dose of 12 mg using the same procedures; if second dose does not terminate the arrhythmia or cause AV block, may administer a third dose of 12 or 18 mg using the same procedures (Ref).

Central line administration: Initial dose should be reduced to 3 mg with subsequent doses of 6 mg, then 9 mg, if needed (Ref).

Heart transplant patients: Initial dose should be reduced to 1 mg; may increase subsequent doses up to 3 mg, if needed (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. Adenosine is not renally eliminated.

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling. Adenosine is not hepatically eliminated.

Dosing: Older Adult

Refer to adult dosing. Older patients may be more sensitive to effects of adenosine.

Dosing: Pediatric

(For additional information see "Adenosine: Pediatric drug information")

Supraventricular tachycardia

Supraventricular tachycardia: Adenocard:

Hemodynamically unstable:

Infants, Children, and Adolescents: Rapid IV, Intraosseous: Initial: 0.1 mg/kg (maximum initial dose: 6 mg/dose); if not effective, increase to 0.2 mg/kg (maximum dose: 12 mg/dose); follow each bolus with NS flush (Ref).

Hemodynamically stable:

Infants, Children, and Adolescents <50 kg: Rapid IV: Initial dose: 0.05 to 0.1 mg/kg via peripheral or central line; maximum initial dose: 6 mg/dose; if not effective within 1 to 2 minutes, increase dose by 0.05 to 0.1 mg/kg increments every 1 to 2 minutes to a maximum single dose of 0.3 mg/kg or 12 mg (whichever is less) or until termination of paroxysmal supraventricular tachycardia (PSVT); follow each bolus with NS flush.

Heart transplant patients: Limited data available: Infants ≥6 months, Children, and Adolescents <50 kg: Rapid IV: Initial dose: 0.025 mg/kg; repeat dosing with gradual dose escalation has been reported (Ref).

Children and Adolescents ≥50 kg: Rapid IV: Initial: 6 mg via peripheral line, if not effective within 1 to 2 minutes, 12 mg may be given; may repeat 12 mg bolus if needed; follow each bolus with NS flush. **Note:** In adults, lower initial doses of 3 mg have been suggested for administration via central line (Ref).

Heart transplant patients: Limited data available: Rapid IV: Initial dose: 0.025 mg/kg (maximum initial dose: 1.5 mg/dose); repeat dosing with gradual dose escalation has been reported (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling. However, adenosine is not renally eliminated.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling. However, adenosine is not hepatically eliminated.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified.

>10%:

Cardiovascular: Cardiac arrhythmia (during cardioversion: 55%; including atrial fibrillation, atrioventricular block, palpitations, premature atrial contractions, premature ventricular contractions, sinus bradycardia, sinus tachycardia), facial flushing (18%)

Respiratory: Dyspnea (12%)

1% to 10%:

Cardiovascular: Chest pressure (7%)

Gastrointestinal: Nausea (3%)

Nervous system: Dizziness (1% to 2%), headache (2%), numbness (1%), tingling of the arms (1%)

Frequency not defined: Respiratory: Hyperventilation

Postmarketing:

Cardiovascular: Asystole (including prolonged) (Tachakra 1992), bradycardia, chest pain (Ismail 2012), coronary artery vasospasm (Ismail 2012), depression of ST segment on ECG (Hage 2009), exacerbation of cardiac arrhythmia (Singh 2015), hypotension (Tachakra 1992), increased ST segment on ECG (Arora 2014), torsades de pointes (Tachakra 1992), transient increase of blood pressure, ventricular fibrillation (Rajkumar 2017), ventricular tachycardia (including unsustained polymorphic ventricular tachycardia) (Romer 1994)

Nervous system: Loss of consciousness, seizure (including tonic-clonic seizure)

Respiratory: Bronchospasm

Contraindications

Hypersensitivity to adenosine or any component of the formulation; second- or third-degree AV block, sick sinus syndrome, or symptomatic bradycardia (except in patients with a functioning artificial pacemaker); known or suspected bronchoconstrictive or bronchospastic lung disease, asthma (manufacturer's labeling); preexcited atrial fibrillation/flutter (ACC/AHA [Joglar 2024]).

Warnings/Precautions

Concerns related to adverse effects:

• Atrial fibrillation/flutter: There have been reports of atrial fibrillation/flutter when administered to patients with paroxysmal supraventricular tachycardia (PSVT) and may be especially problematic in patients with PSVT and underlying Wolff-Parkinson-White syndrome; has also been reported in patients with or without a history of atrial fibrillation undergoing myocardial perfusion imaging with adenosine infusion.

• Cardiovascular events: Cardiac arrest (fatal and nonfatal), myocardial infarction, cerebrovascular accident (hemorrhagic and ischemic), and sustained ventricular tachycardia (requiring resuscitation) have occurred when used for pharmacologic stress testing. Avoid use in patients with signs or symptoms of unstable angina, acute myocardial ischemia, or cardiovascular instability due to possible increased risk of significant cardiovascular consequences. Appropriate measures for resuscitation should be available during use.

• Conduction disturbances: Adenosine decreases conduction through the AV node and may produce first-, second-, or third-degree heart block. Patients with preexisting SA nodal dysfunction may experience prolonged sinus pauses after adenosine; use caution in patients with first-degree atrioventricular (AV) block or bundle branch block; use is contraindicated in patients with high-grade AV block, sinus node dysfunction, or symptomatic bradycardia (unless a functional artificial pacemaker is in place). Rare, prolonged episodes of asystole have been reported, with fatal outcomes in some cases. Discontinue adenosine in any patient who develops persistent or symptomatic high-grade AV block.

• Hypersensitivity: Hypersensitivity reactions (including dyspnea, pharyngeal edema, erythema, flushing, rash, or chest discomfort) have been reported.

• Hypertension: Systolic and diastolic pressure increases have been observed. In most instances, blood pressure increases resolved spontaneously within several minutes; occasionally, hypertension lasted for several hours.

• Hypotension: May produce profound vasodilation with subsequent hypotension. When used as a bolus dose (PSVT), effects are generally self-limiting (due to the short half-life of adenosine). However, when used as a continuous infusion (pharmacologic stress testing), effects may be more pronounced and persistent, corresponding to continued exposure. Use infusions with caution in patients with autonomic dysfunction, carotid stenosis (with cerebrovascular insufficiency), hypovolemia, pericarditis, pleural effusion and/or stenotic valvular heart disease; discontinue infusion in patients who develop persistent or symptomatic hypotension.

• Proarrhythmic effects: Monitor for proarrhythmic effects (eg, polymorphic ventricular tachycardia) during and shortly after administration/termination of arrhythmia. The benign transient occurrence of atrial and ventricular ectopy is common upon termination of arrhythmia.

• Seizures: Seizures (new-onset or recurrent) have been reported; risk may be increased with concurrent use of aminophylline. Concomitant use of any methylxanthine (eg, aminophylline, caffeine, theophylline) is not recommended.

Disease-related concerns:

• Arrhythmia (wide-complex tachycardia): Avoid use in irregular or polymorphic wide-complex tachycardias; may cause degeneration to ventricular fibrillation (AHA [Neumar 2010]).

• Heart transplant recipients: Use with extreme caution in heart transplant recipients; adenosine may cause prolonged asystole; reduction of initial adenosine dose is recommended (AHA [Wigginton 2025]); considered by some to be contraindicated in this setting (Delacrétaz 2006).

• Pulmonary artery hypertension: Acute vasodilator testing (not an approved use): Use with extreme caution in patients with concomitant heart failure (LV systolic dysfunction with significantly elevated left heart filling pressures) or pulmonary veno-occlusive disease/pulmonary capillary hemangiomatosis; significant decompensation has occurred with other highly selective pulmonary vasodilators resulting in acute pulmonary edema.

• Respiratory disease: Avoid use in patients with bronchoconstriction or bronchospasm (eg, asthma); dyspnea, bronchoconstriction, and respiratory compromise have occurred during use. Use caution in patients with obstructive lung disease not associated with bronchoconstriction (eg, emphysema, bronchitis). Immediately discontinue therapy if severe respiratory difficulty is observed. Appropriate measures for resuscitation should be available during use.

Concurrent drug therapy issues:

• Caffeine: Pharmacologic stress testing: Since caffeine antagonizes the activity of adenosine, withhold for 5 half-lives prior to adenosine use; avoid dietary caffeine for at least 12 hours prior to pharmacologic stress testing (ASNC [Henzlova 2016]).

• Dipyridamole: Concomitant use potentiates the effects of adenosine; reduction of initial adenosine dose is recommended when used for SVT (AHA [Neumar 2010]); withhold dipyridamole-containing medications for at least 48 hours prior to pharmacologic stress testing (ASNC [Henzlova 2016]).

• Theophylline or aminophylline: Pharmacologic stress testing: Since methylxanthines antagonizes the activity of adenosine, withhold for 5 half-lives prior to adenosine use whenever possible.

Special populations:

• Older adult: Use with caution in older patients; may be at increased risk of hemodynamic effects, bradycardia, and/or AV block.

Dosage form specific issues:

• Atrioventricular block: Transient AV block is expected when administered as an IV bolus for treatment of paroxysmal supraventricular tachycardia (PSVT). When used for PSVT, at the time of conversion to normal sinus rhythm, a variety of new rhythms may appear on the ECG. Administer as a rapid bolus, either directly into a vein or (if administered into an IV line), as close to the patient as possible (followed by saline flush). Dose reduction recommended when administered via central line (AHA [Wigginton 2025]).

Other warnings/precautions:

• Appropriate use: ECG monitoring is required during use. Equipment for resuscitation and trained personnel experienced in handling medical emergencies should always be immediately available. Adenosine does not convert atrial fibrillation/flutter to normal sinus rhythm; however, may be used diagnostically in these settings if the underlying rhythm is not apparent.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous:

Generic: 3 mg/mL (20 mL, 30 mL); 6 mg/2 mL (2 mL); 12 mg/4 mL (4 mL)

Solution, Intravenous [preservative free]:

Generic: 3 mg/mL (20 mL, 30 mL); 6 mg/2 mL (2 mL); 12 mg/4 mL (4 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Adenosine (Diagnostic) Intravenous)

3 mg/mL (per mL): $6.29 - $7.80

Solution (Adenosine Intravenous)

6 mg/2 mL (per mL): $3.60 - $9.60

12 mg/4 mL (per mL): $3.00 - $7.80

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous:

Generic: 3 mg/mL (2 mL, 4 mL); 6 mg/2 mL (2 mL); 12 mg/4 mL (4 mL)

Administration: Adult

IV: For rapid IV bolus use; administer IV push over 1 to 2 seconds at a peripheral IV site as proximal as possible to trunk (not in lower arm, hand, lower leg, or foot); immediately after each bolus, administer an NS flush (eg, 20 mL (Ref); refer to institutional policies and procedures). Use of 2 syringes (one with adenosine dose and the other with NS flush) connected to a T-connector or stopcock is recommended. Alternatively, if no stopcock is available or the patient only has a single-port IV, prepare a single syringe with adenosine 6 mg and 18 mL of NS and administer at a peripheral IV site (Ref).

Fractional flow reserve testing (off-label use): Intracoronary: Administer rapidly followed by a 10 mL NS flush (Ref).

For IV continuous infusion: Only administer via a peripheral line.

Preparation for Administration: Adult

Parenteral: IV: Doses ≥0.6 mg: Give undiluted.

Administration: Pediatric

Parenteral: Adenocard: For rapid bolus IV use, administer over 1 to 2 seconds at peripheral IV site closest to patient's heart (IV administration into lower extremities may result in therapeutic failure or requirement of higher doses); follow each bolus with NS flush (infants and children: 5 to 10 mL; adults: 20 mL); **Note:** The use of 2 syringes (one with adenosine dose and the other with NS flush) connected to a T-connector or stopcock is recommended for IV and intraosseous administration (Ref).

Preparation for Administration: Pediatric

Parenteral: IV:

Doses ≥600 mcg: Give undiluted.

Doses <600 mcg: Further dilution of dose may be necessary to ensure complete and accurate administration; dilution with NS to a final concentration of 300 to 1,000 mcg/mL has been used; to prepare a 300 mcg/mL solution, add 3 mg of adenosine (1 mL) to 9 mL of NS; to prepare a 1,000 mcg/mL, add 3 mg of adenosine (1 mL) to 2 mL of NS (Ref).

Storage/Stability

Store at 20°C to 25°C (68°F to 77°F); excursions permitted between 15°C and 30°C (59°F and 86°F). Do **not** refrigerate; crystallization may occur (may dissolve by warming to room temperature).

Use: Labeled Indications

Paroxysmal supraventricular tachycardia: Treatment of paroxysmal supraventricular tachycardia; when clinically advisable, appropriate vagal maneuvers should be attempted prior to adenosine administration. **Note:** Not effective for conversion of atrial fibrillation, atrial flutter, or ventricular tachycardia.

Pharmacologic cardiac stress testing, diagnostic aid: Pharmacologic stress agent used in myocardial perfusion thallium-201 scintigraphy.

Use: Off-Label: Adult

Fractional flow reserve testing, diagnostic aid; Tachyarrhythmia, diagnostic aid in hemodynamically stable patients

Medication Safety Issues

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (antiarrhythmic agent, IV) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Bradycardia-Causing Agents: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Caffeine and Caffeine Containing Products: May decrease therapeutic effects of Adenosine. Management: Monitor for decreased effect of adenosine if patient is receiving caffeine; significantly higher adenosine doses or alternative agents may be required. Discontinue caffeine 24 hours in advance of scheduled diagnostic use of adenosine if possible. *Risk D: Consider Therapy Modification*

CarBAMazepine: May increase adverse/toxic effects of Adenosine. Specifically, the risk of higher degree heart block may be increased. Management: Monitor for increased degrees of heart block and bradycardia when these agents are combined. Consider using a lower initial dose of adenosine (3 mg) for the treatment of supraventricular tachycardia in patients who are receiving carbamazepine. *Risk C: Monitor*

Ceritinib: Bradycardia-Causing Agents may increase bradycardic effects of Ceritinib. Management: If this combination cannot be avoided, monitor patients for evidence of symptomatic bradycardia, and closely monitor blood pressure and heart rate during therapy. *Risk D: Consider Therapy Modification*

Digoxin: May increase adverse/toxic effects of Adenosine. *Risk C: Monitor*

Dipyridamole: May increase adverse/toxic effects of Adenosine. Specifically, cardiovascular effects of adenosine may be enhanced. Adenosine dose reduction may be needed. Management: For patients requiring pharmacologic stress testing with adenosine, hold dipyridamole tablets for 48 hours. Hold aspirin/dipyridamole capsules for 24 to 48 hours. For treatment of SVT, monitor for prolonged adenosine effects, consider lower initial doses. *Risk D: Consider Therapy Modification*

Etrasimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Fexinidazole: Bradycardia-Causing Agents may increase arrhythmogenic effects of Fexinidazole. *Risk X: Avoid*

Fingolimod: Bradycardia-Causing Agents may increase bradycardic effects of Fingolimod. Management: Consult with the prescriber of any bradycardia-causing agent to see if the agent could be switched to an agent that does not cause bradycardia prior to initiating fingolimod. If combined, perform continuous ECG monitoring after the first fingolimod dose. *Risk D: Consider Therapy Modification*

Ivabradine: Bradycardia-Causing Agents may increase bradycardic effects of Ivabradine. *Risk C: Monitor*

Lacosamide: Bradycardia-Causing Agents may increase AV-blocking effects of Lacosamide. *Risk C: Monitor*

Landiolol: Bradycardia-Causing Agents may increase bradycardic effects of Landiolol. *Risk X: Avoid*

Midodrine: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Nicotine: May increase AV-blocking effects of Adenosine. Nicotine may increase tachycardic effects of Adenosine. *Risk C: Monitor*

Ozanimod: May increase bradycardic effects of Bradycardia-Causing Agents. *Risk C: Monitor*

Ponesimod: Bradycardia-Causing Agents may increase bradycardic effects of Ponesimod. Management: Avoid coadministration of ponesimod with drugs that may cause bradycardia when possible. If combined, monitor heart rate closely and consider obtaining a cardiology consult. Do not initiate ponesimod in patients on beta-blockers if HR is less than 55 bpm. *Risk D: Consider Therapy Modification*

Siponimod: Bradycardia-Causing Agents may increase bradycardic effects of Siponimod. Management: Avoid coadministration of siponimod with drugs that may cause bradycardia. If combined, consider obtaining a cardiology consult regarding patient monitoring. *Risk D: Consider Therapy Modification*

Theophylline Derivatives: May decrease therapeutic effects of Adenosine. Management: Consider alternatives to this combination if possible. Theophylline may decrease adenosine efficacy and higher adenosine doses may be required. When using adenosine for diagnostic studies, discontinue theophylline derivatives 5 half-lives prior to test. *Risk D: Consider Therapy Modification*

Pregnancy Considerations

Animal reproduction studies have not been conducted. Adenosine is an endogenous substance and adverse fetal effects would not be anticipated. Adenosine is recommended for the acute treatment of SVT in pregnant women. The usual recommended doses may be used, although higher doses may be needed in some cases (ACC/AHA/HRS [Page 2016]). AHA guidelines suggest use is safe and effective in pregnancy (AHA [Neumar 2010]).

Breastfeeding Considerations

It is not known if adenosine is excreted in breast milk following maternal administration. Adenosine is endogenous in breast milk (Sugawara 1995). Due to the potential for adverse reactions in the nursing infant, the manufacturer recommends a decision be made to interrupt nursing or not administer adenosine taking into account the importance of treatment to the mother.

Dietary Considerations

Avoid dietary caffeine for at least 12 hours prior to pharmacologic stress testing.

Monitoring Parameters

ECG, heart rate, blood pressure; consult individual institutional policies and procedures

Mechanism of Action

Antiarrhythmic actions: Slows conduction time through the AV node, interrupting the re-entry pathways through the AV node, restoring normal sinus rhythm

Myocardial perfusion scintigraphy: Aԁenosine also causes coronary vasodilation and increases blood flow in normal coronary arteries with little to no increase in stenotic coronary arteries; thallium-201 uptake into the stenotic coronary arteries will be less than that of normal coronary arteries revealing areas of insufficient blood flow.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Rapid

Duration: Very brief

Metabolism: Removed from systemic circulation primarily by vascular endothelial cells and erythrocytes (by cellular uptake); rapidly metabolized intracellularly; phosphorylated by aԁenosine kinase to aԁenosine monophosphate (AMP) which is then incorporated into high-energy pool; intracellular aԁenosine is also deaminated by aԁenosine deaminase to inosine; inosine can be metabolized to hypoxanthine, then xanthine and finally to uric acid.

Half-life elimination: <10 seconds

Patient Counseling Points

(For additional information see ”Aԁenosine: Patient drug information”)

What is this drug used for?

• It is used to treat certain types of abnormal heartbeats.

• It is used during a stress test of the heart.

• It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

• Dizziness or headache

• Flushing

• Stomach pain

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• High or low blood pressure like very bad headache or dizziness, passing out, or change in eyesight

• Weakness on 1 side of the body, trouble speaking or thinking, change in balance, drooping on one side of the face, or blurred eyesight

• Fast, slow, or abnormal heartbeat

• Chest pain or pressure

• Severe dizziness or passing out

• Shortness of breath

• Seizures

• Throat, neck, or jaw pain

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

Find brand name(s) by country or region (for United States and Canada see separate Brand Names sections)

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Adenoject;
- (AR) Argentina: Adenosina biol | Euritsin;
- (AT) Austria: Adenosin baxter | Adenosin ebewe | Adenosin hikma | Adenosin panpharma;
- (AU) Australia: Adenocor | Adenoscan | Adenosine baxter | Adenosine juno | Adenosine viatris | Adsine | Aspen Adenosine;
- (BD) Bangladesh: Adecard;
- (BE) Belgium: Adenocor;
- (BG) Bulgaria: Adenocor;
- (BR) Brazil: Adenosina | Lowe;
- (CH) Switzerland: Krenosin;
- (CL) Chile: Adenosina | Cardosine | Tricor;
- (CN) China: Adenosine | Adenosine injection for diagnosis;
- (CO) Colombia: Adenocor | Adenosina | Adenotroy;
- (CR) Costa Rica: Adenocor | Adenosina biol;
- (CZ) Czech Republic: Adenocor | Adenosin;
- (DE) Germany: Adenoscan | Adenosin accord | Adenosin altamedics | Adenosin baxter | Adenosin item | Adenosin life medical | Adenosin rotexmedica | Adrekar | Rotop Adenosin;
- (EC) Ecuador: Adenocor | Adenosina;
- (EE) Estonia: Adenocor | Adenosin life medical;
- (ES) Spain: Adenocor | Adenoscan | Adenosina accord | Adenosina hikma;
- (FI) Finland: Adenocor | Adenoscan | Adenosin life medical;
- (FR) France: Adenoscan | Adenosine accord | Adenosine hikma | Adenosine medac | Krenosin | Striadyne;
- (GB) United Kingdom: Adenocor | Adenoscan | Adenosine focus | Adenotriphos | Adenyl;
- (GR) Greece: Adenocor | Adenorythm;
- (HK) Hong Kong: Adenoscan;
- (IE) Ireland: Adenocor;
- (IL) Israel: Adenocor;
- (IN) India: Adenocor | Adenoz;
- (IT) Italy: Adenoscan | Adenosina accord | Adenosina hikma | Krenosin;
- (JO) Jordan: Xoria;
- (JP) Japan: Adetphos | Adetphos kowa;
- (KR) Korea, Republic of: Adenocor | Anosin | Denosine;
- (KW) Kuwait: Adenocor | Adenorythm;
- (LB) Lebanon: Adenocor;

- (LT) Lithuania: Adenocor | Adenosina | Adenoz | Adrekar;
- (LV) Latvia: Adenosin life medical;
- (MX) Mexico: Alvolden | Narodex;
- (MY) Malaysia: Adenocor | Adenorythm | Myodeen;
- (NL) Netherlands: Adenocor | Adenosine eureco pharma | Adenosine hikma;
- (NO) Norway: Adenocor | Adenosin | Adenosin item | Adenosin life medical;
- (NZ) New Zealand: Adenocor | Adenoscan | Adenosine claris | Adenosine focus;
- (PA) Panama: Adenocor | Caden;
- (PE) Peru: Adenocor | Adenosina | Adenoz | Cardimax;
- (PH) Philippines: Adesan | Cardiovert;
- (PL) Poland: Adenocor | Adenorythm | Adenoscan | Adenosine delfarm | Soladen;
- (PR) Puerto Rico: Adenocard | Adenoscan | Adenosine;
- (PT) Portugal: Adenocor | Adenoscan | Caden;
- (PY) Paraguay: Adenosina biol | Adenosina gp pharm;
- (QA) Qatar: Adenocor | Cardesine | Krenosin;
- (RO) Romania: Adenocor | Adenosin life medical | Adozin;
- (RU) Russian Federation: Adenosintriphosphate sodium | Adenosintriphosphate sodium vial | Sodium adenosintriphosphate darnitsa;
- (SA) Saudi Arabia: Adenocor | Adenohardt | Caden;
- (SE) Sweden: Adenoscan | Adenosin life medical;
- (SG) Singapore: Adenocor;
- (SI) Slovenia: Adenocor | Adenosin altamedics | Adenosin hikma | Adrekar;
- (TH) Thailand: Adenocor;
- (TR) Turkey: Adenosin-L.M. | Adenotek | Adozin;
- (TW) Taiwan: Adenocor | Adenozer;
- (UY) Uruguay: Adenocor | Cardenosina;
- (ZA) South Africa: Adenocor | Adenosin life medical

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